

Impact Statement: Optimizing RAG Document Retrieval for Agronomic Advice

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Purpose and Positive Impacts

This project develops a hybrid document retrieval architecture designed to accurately fetch agricultural extension literature based on inquiries from smallholder farmers in sub-Saharan Africa. By bridging the vocabulary gap between a farmer's informal description of crop symptoms and the formal scientific terminology used in agronomic manuals, the system optimizes the foundational evidence layer for downstream Retrieval-Augmented Generation (RAG). The primary benefit of this system is empowering agricultural extension workers with rapid, highly relevant evidence to provide safe and timely recommendations. Ultimately, this supports community well-being by improving crop yields, preventing the misapplication of toxic fertilizers, and enhancing regional food security.

Potential Risks and Harms

Despite these positive impacts, deploying this technology carries notable societal and structural risks. A significant concern is automation bias, where users might develop an undue trust in the retrieved results and fail to exercise their independent domain expertise. If the system misunderstands local dialects or symptom descriptions, it may retrieve irrelevant documents, leading to poor downstream advice that could result in crop failure, lost income, or environmental damage. Furthermore, the system could exacerbate existing inequalities if it performs highly on well-documented staple crops but fails to support farmers cultivating minor regional crops.

Values

Our design promotes inclusivity and fairness by integrating dense semantic search, ensuring farmers lacking formal scientific education can still receive accurate guidance based on conversational symptom descriptions. The project heavily champions transparency and accountability by returning verifiable source document identifiers, allowing users to trace and verify the original evidence.

Mitigation Strategies

To mitigate identified risks, our team frames this system strictly as a decision-support tool designed to augment, rather than replace, human agricultural judgment. We counteract representation bias by rigorously validating performance across

underrepresented crops and minor pests. To ensure safety, we employ customized lexical search baselines to prioritize exact matches for regulated chemical dosages and treatments.

Limitations and Uncertainties

Foreseeing all the long-term societal impacts of this research involves inherent uncertainties. A primary limitation is our reliance on a static corpus of 695 documents, which cannot fully represent the diverse linguistic patterns and specialized agricultural contexts across all African farming communities. Additionally, it remains uncertain whether real-world users will consistently verify the retrieved documents in high-stress scenarios, emphasizing the need for robust socio-technical evaluation by agricultural experts prior to deployment.